

Lesson 8: Our Strategy, End to End

Analyst onboarding · Lesson 8 of 10 · L/S Gamma Desk, QUANTT

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Goal: know every rule our desk trades on and how a day runs, from data download to hedge.

You need: Lessons 1–7.

1 The idea in one line

Every day, forecast SPY's volatility to expiry. If options are **too cheap**, buy one and scalp it (long gamma). If they are **too expensive**, sell one and scalp it (short gamma). Otherwise **do nothing**.

2 Rule 1: the signal

$$VRP = IV_{ATM} - RV_{forecast}$$

VRP	Signal	Trade
above +0.02	short gamma	Sell 1 ATM put, delta hedge
below −0.05	long gamma	Buy 1 ATM call, delta hedge
in between	flat	No new trade

Why the band is lopsided: the VRP means IV is usually above RV (Lesson 6). Buying volatility fights that premium, so we demand a bigger edge before buying (IV at least 5 vol points *below* our forecast) than before selling (IV at least 2 points *above* it). The **flat zone** in between keeps us out when the forecast's edge is too small to beat costs and forecast error.

3 Rule 2: which option

- **Expiry:** the Friday expiry closest to **30 days**, within 21–45 DTE. One-to-two month options give the best delta-hedged Sharpe ratio (Sepp 2013), and very short-dated gamma swings too fast for once-a-day hedging.
- **Strike:** the one closest to SPY's price (**ATM**), where gamma, vega and the VRP are largest.
- **Calls for long gamma, puts for short gamma.** Once hedged, they're nearly the same bet (Lesson 4), but buying calls avoids paying the put skew, and selling puts collects it.

4 Rule 3: size

1 contract per trade, at most 1 open position. Paper trading is first a test of the machinery. We size up only once we have a P&L history to measure risk with.

5 Rule 4: hedging

Once a day, trade SPY shares to bring the total delta back to zero (Lesson 5).

6 Rule 5: exits (the first trigger to fire wins)

Order	Trigger	Rule
1	time	7 or fewer days to expiry: avoid the last week's gamma and theta cliff
2	stop	Short put: loss $\geq 1 \times$ premium received. Long call: loss $\geq 50\%$ of premium paid. Measured on option P&L plus hedge P&L
3	profit	Short put: profit $\geq 50\%$ of premium received
4	flip	The signal turns to the opposite side: close, then open the other side
5	flat	The signal has been flat for 3 days in a row

Every number above lives in `configs/paper.toml`, so changing a rule never needs a code change.

7 Other structures we could trade

Long gamma	Short gamma	Trade-off
Long call / put	Short call / put	Simplest, one leg. Starts with a big delta to hedge
Long straddle (call + put, same strike)	Short straddle	Starts delta neutral, most gamma. Most expensive / most risk
Long strangle (OTM call + put)	Short strangle	Cheaper, less gamma near the money
Reverse calendar	Iron condor (strangle + bought wings)	Less vega / capped loss, but more legs and costs

All ten have execution code ready in `core/Algos/` (folders `LG_*` and `SG_*`). For now we deliberately use only the simplest pair. Choosing between structures by market regime is future work (Lesson 10).

8 A trading day, step by step (about 10:15 ET)

1. **Data:** the pipeline downloads SPY prices and VIX indexes (yfinance) plus the SPY option chain (IBKR, delayed quotes).
2. **Forecast:** fit GARCH on SPY returns and forecast volatility to the target expiry.
3. **Signal:** compare the forecast with that expiry's ATM IV, then log it.
4. **Exits:** if a position is open, mark it at the mid, compute option + hedge P&L, and check the five triggers.
5. **Entry:** if no position is open and the signal isn't flat, open the chosen option at mid + \$0.05.
6. **Hedge:** trade shares to make delta zero.
7. **Log:** everything goes to `results/paper/` (`signals`, `trades`, `hedges`, `position.json`).

Run it with `python -m lsgamma.trading.runner` (add `--dry-run` to see decisions without trading). All commands are in `docs/scripts.md`.

9 The code map

Folder	Job
pipeline/ forecasting/ signals/ trading/ core/Algos/	Daily data snapshot (yfinance + IBKR chain) GARCH, HAR (FORECASTERS registry) vrp_band (SIGNALS registry) Selection, exits, P&L, ledger, runner, smoke test Execution only: <code>broker.py</code> , the 10 structures, Hedging/
configs/ tests/	<code>paper.toml</code> : every number Offline tests with a fake broker (<code>pytest</code>)

Design rule: the algos only *describe* trades. Every decision lives in `signals/` and `trading/`. That's why new signals or hedgers (e.g. an RL hedger) can be swapped in without touching execution.

10 Safety rules

- **Paper only.** `broker.connect()` refuses any account that isn't a paper account (IDs starting with "DU").
- **Never commit data or keys.** Market data from the club's R2 bucket and WRDS is licensed to Queen's. `.env`, `data/` and `results/` are never committed.

11 Words to know

- **Signal band:** the thresholds that turn the VRP into long, short or flat.
- **Exit trigger:** a rule that closes a position.
- **Dry run:** a run that decides but doesn't trade.
- **Registry:** a dictionary mapping names to interchangeable models.

12 Check yourself

1. Forecast 16%, ATM IV 13%. What does the desk do?
2. A short put opened for \$11.20 now shows $-\$1,150$ of combined P&L. Which trigger fires?
3. Where would you change the long band from -0.05 to -0.04 ?

Answers: (1) $VRP = 13\% - 16\% = -0.03$ is inside the band: flat, no trade. (2) The stop: the loss exceeds $1\times$ the \$1,120 premium. (3) `configs/paper.toml`, `long_band`.